Bioinformatics

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Functional enrichment analysis and enrichR

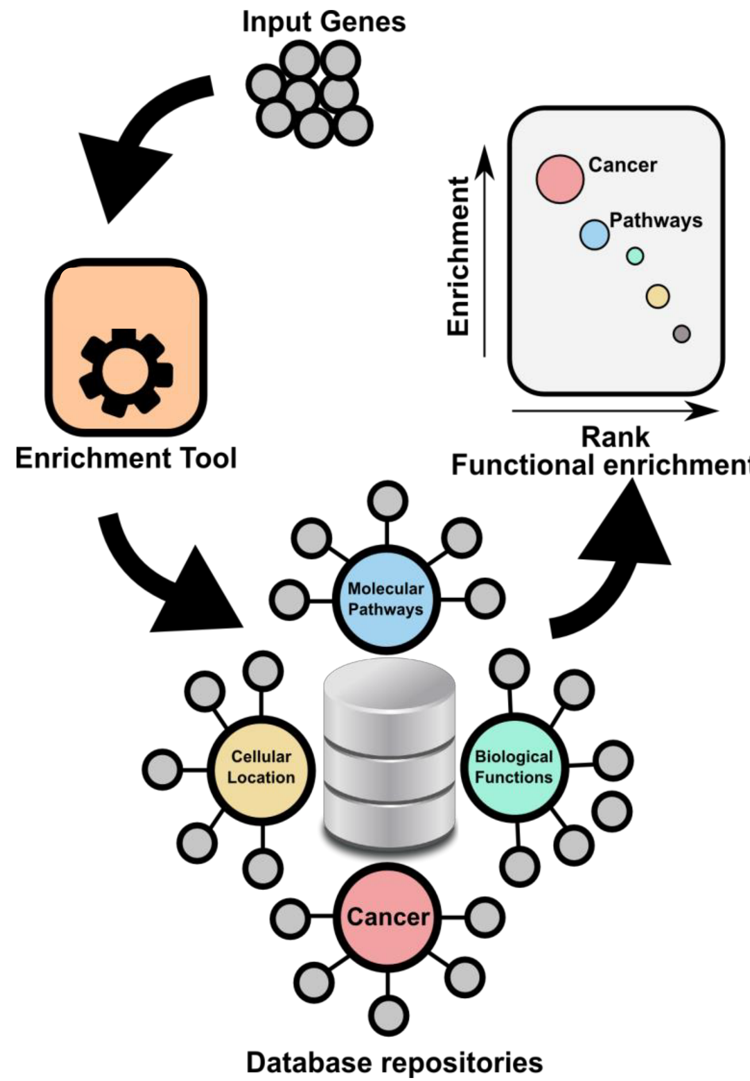
Functional enrichment

Functional enrichments

- The existing technologies allow to simultaneously highlight the activity of thousands of genes. Results of these analyses are candidate genes that may be functionally relevant in the examined experimental condition

- The interpretation of such results requires the annotation of the selected genes to functional and structural categories, using appropriate controlled vocabularies, that univocally define their features enabling a standardized analysis

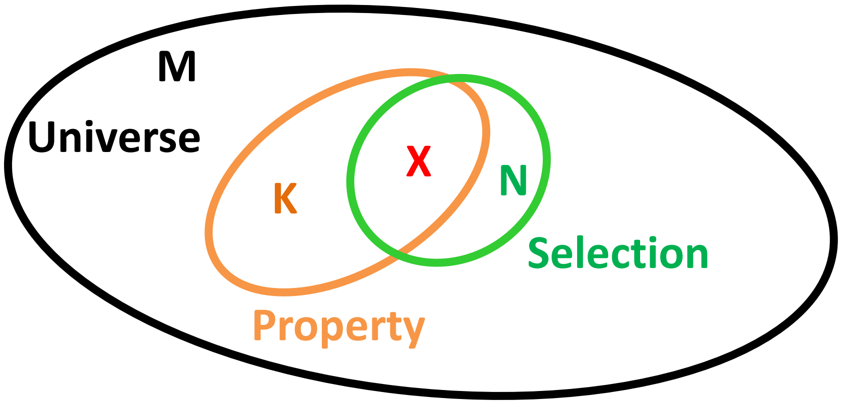
- Using these annotations, we can group the genes of a given list according to their functional and structural features, and study the statistical significance of these categories



Hypergeometric test

Test of *over*-representation: ***which are the significant categories in the list of selected genes?***

- **M**: universe (human genome)
- **K**: set of elements with a given property (all genes annotated for a given functional category)
- **N**: set of elements for which you want to measure the statistical significance of the enrichment in the property K (the list of selected genes)

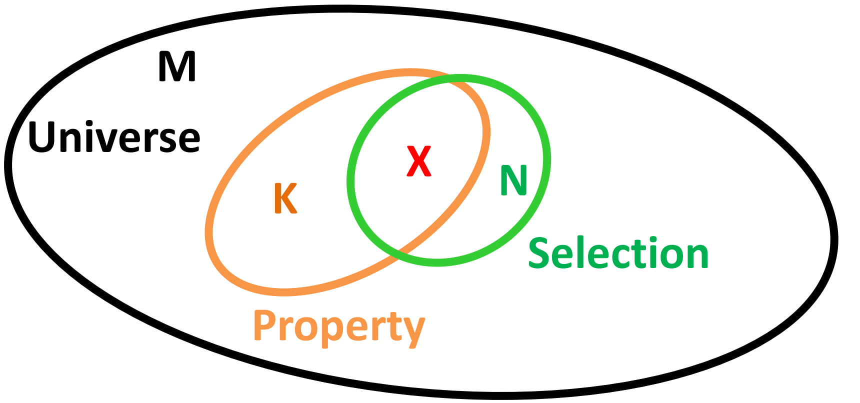


- **H_0:** if a gene is annotated for a particular category and belongs to the list of selected genes is due to chance
- **H_1**: the class of selected genes in the tested list is enriched in the considered category. This means that the category is over-represented on the list of selected genes
- p-value -> 0 The null hypothesis is rejected and we can conclude that the list of genes is enriched for the considered functional category respect to what one would expect by chance
- p-value -> 1 the null hypothesis is accepted, that is list does not enriched for the considered functional category

Hypergeometric test

Probability of observing X elements with the K property is given by:

$$P(X) = \frac{\binom{K}{X}\binom{M-K}{N-X}}{\binom{M}{N}}$$



$\binom{K}{X}$ number of possible draws of X objects from K objects

$\binom{M-K}{N-X}$ number of possible draws of the remaining N-X objects among the M-K objects

$\binom{M}{N}$ number of possible draws of N objects from M objects

The hypergeometric distribution is a **discrete probability distribution** that calculates the likelihood an event happens X times in N trials when you are sampling from a small population **without replacement**, so describes the probability of successes in draws without replacement

Hypergeometric test

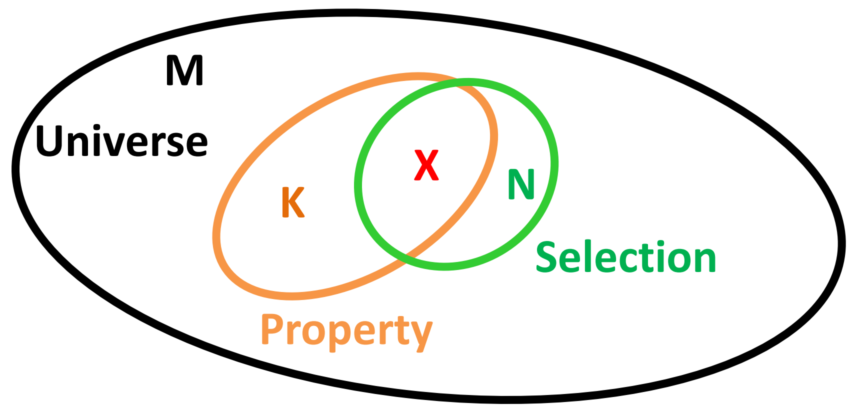
Probability of observing X elements with the K property is given by:

# of successes **# of unsuccesses**

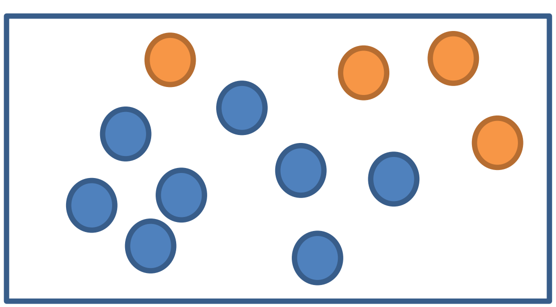
$$P(X) = \frac{\binom{K}{X}\binom{M-K}{N-X}}{\binom{M}{N}}$$

Total draws

Given an urn with two colors of marbles, (M-K) blue and (K) orange. The hypergeo test computes the probability of extracting X number of orange marbles by drawing without replacement N marbles from the urn



Example:
M = # of total marbles (12)
K = # of orange marbles (4)
M-K = # of blue marbles (8)
N = # of drawn marbles



Example:
M = transcriptome (all human genes)
N = # of selected genes (e.g., DEG)
K = # of genes annotated for a functional category your are testing (e.g., cell cycle)
X = # of genes of your list annotated for cell cycle

The ratio X/N is greater than K/M?

The list of N genes is over-represented in that category

Hypergeometric test

How many beans are in the box? $M = 10$

How many beans are chosen, at random, without replacement? $N = 5$

How many yucky flavors are in the box? $K = 4$

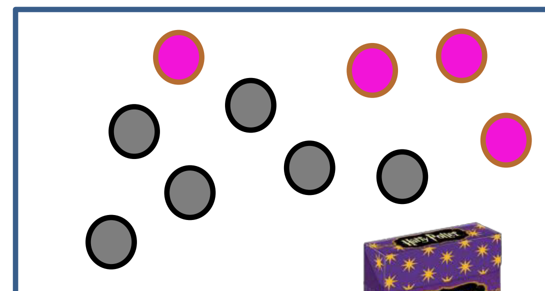
How many yucky flavors to be chosen? $X = 2$

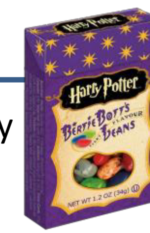
Yucky

Tasty

$$P(X) = \frac{\binom{K}{X}\binom{M-K}{N-X}}{\binom{M}{N}}$$

Total

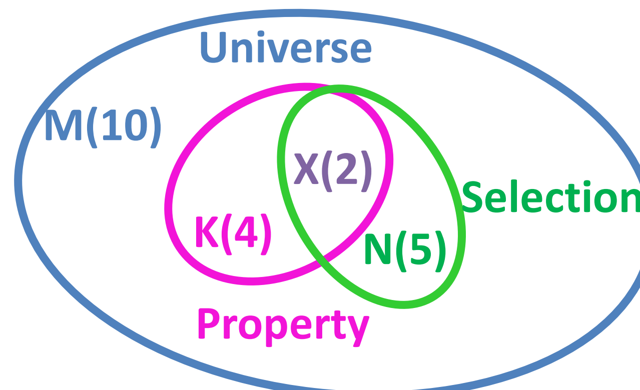




bertie bott's every flavor beans

Probability of extracting 2 (out of 4) yucky beans by drawing without replacement 5 beans

$$P(X) = \frac{\binom{4}{2}\binom{10-4}{5-2}}{\binom{10}{5}}$$



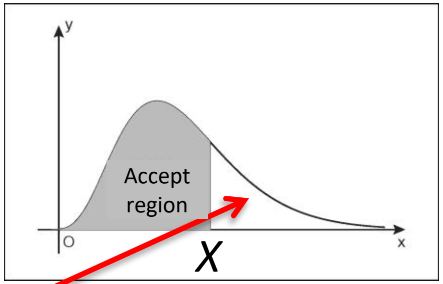
$$P(X) = \frac{\binom{4}{2}\binom{6}{3}}{\binom{10}{5}} = \frac{\left(\frac{4!}{2!(4-2)!}\right)\left(\frac{6!}{3!(6-3)!}\right)}{\left(\frac{10!}{5!(10-5)!}\right)} = \frac{(6)(20)}{252} = 0.476$$

Hypergeometric test

Test of *over*-representation: ***which are the significant categories in the list of selected genes?***

Probability of observing X or more than elements with the K property is given by:

$$\Pr = 1 - \sum_{i=0}^{X-1} \frac{\binom{K}{i}\binom{M-K}{N-i}}{\binom{M}{N}} = \boxed{\sum_{i=X}^{N} \frac{\binom{K}{i}\binom{M-K}{N-i}}{\binom{M}{N}}}$$



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Hypergeometric test

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$$\Pr = 1 - \sum_{i=0}^{X-1} \frac{\binom{K}{i}\binom{M-K}{N-i}}{\binom{M}{N}} = \sum_{i=X}^{N} \boxed{\frac{\binom{K}{i}\binom{M-K}{N-i}}{\binom{M}{N}}} \rightarrow$$

Hypergeometric distribution (probability density)

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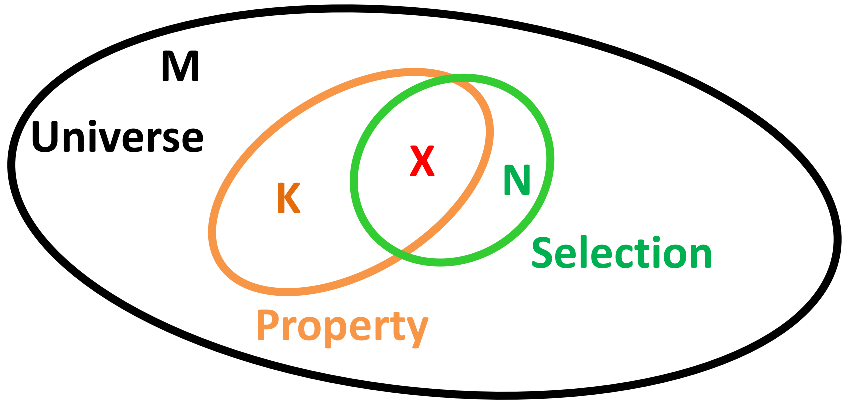

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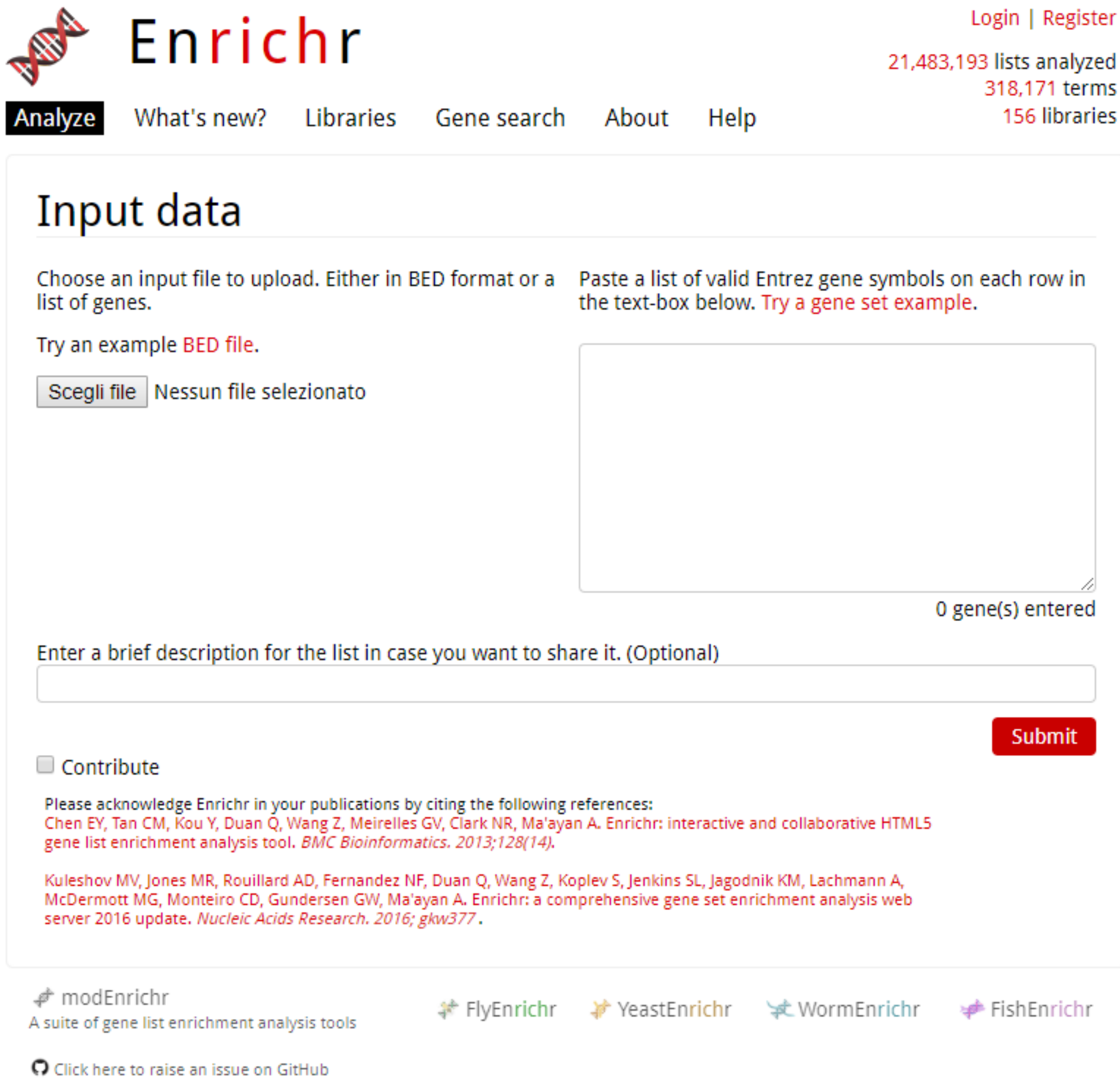
- p-value -> 1 the null hypothesis is accepted, that is list does not enriched for the considered functional category

enrichr

interactive and collaborative gene list
enrichment analysis tool

https://amp.pharm.mssm.edu/Enrichr/


Enrichr
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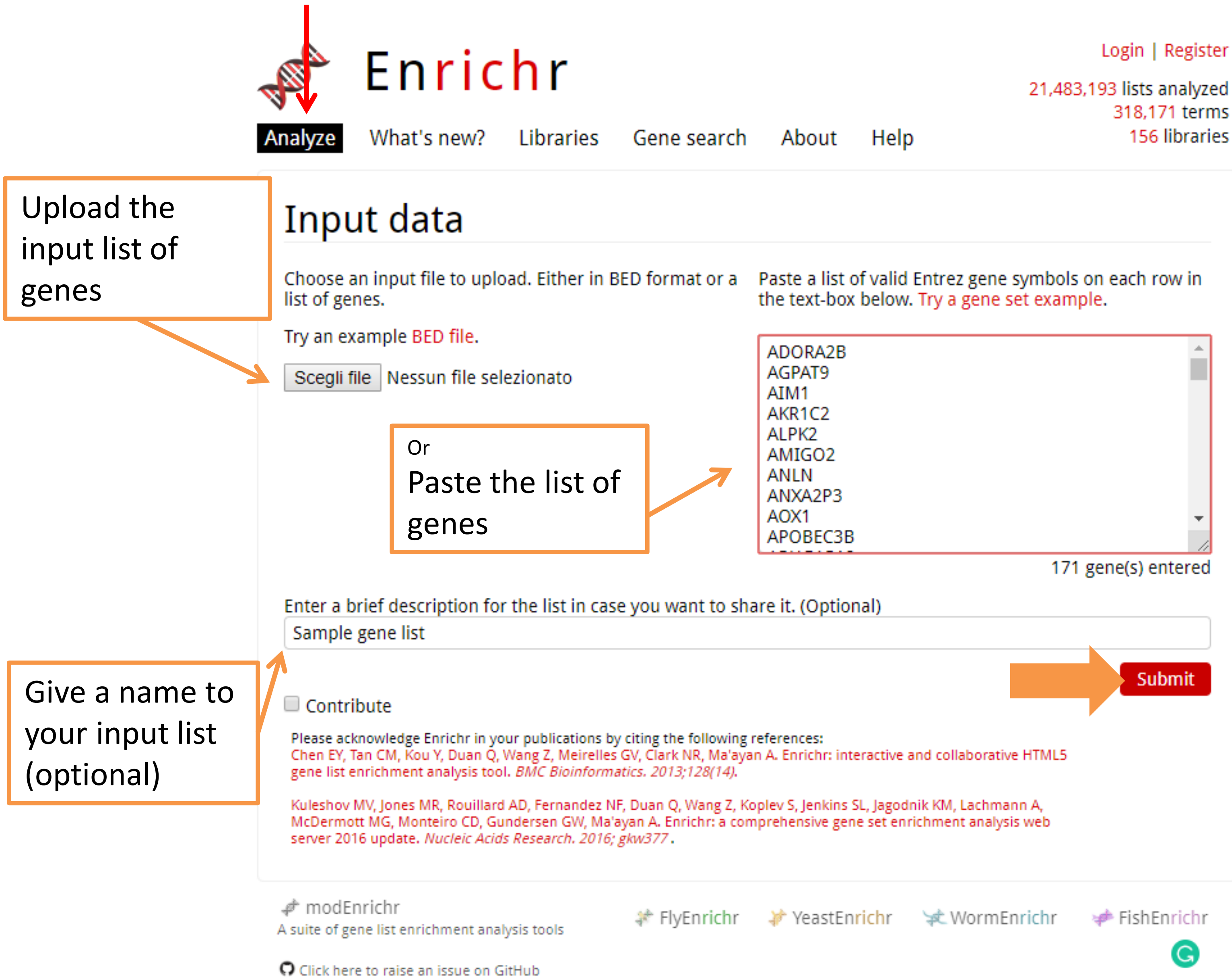
21,483,193 lists analyzed
318,171 terms
156 libraries
Analyze
What's new?
Libraries
Gene search
About
Help
Input data
Choose an input file to upload. Either in BED format or a list of genes.
Try an example BED file.
Scegli file
Nessun file selezionato
Paste a list of valid Entrez gene symbols on each row in the text-box below. Try a gene set example.
0 gene(s) entered
Enter a brief description for the list in case you want to share it. (Optional)
Submit
Contribute
Please acknowledge Enrichr in your publications by citing the following references:
Chen EY, Tan CM, Kou Y, Duan Q, Wang Z, Meirelles GV, Clark NR, Ma'ayan A. Enrichr: interactive and collaborative HTML5 gene list enrichment analysis tool. BMC Bioinformatics. 2013;128(14).
Kuleshov MV, Jones MR, Rouillard AD, Fernandez NF, Duan Q, Wang Z, Koplev S, Jenkins SL, Jagodnik KM, Lachmann A, McDermott MG, Monteiro CD, Gundersen GW, Ma'ayan A. Enrichr: a comprehensive gene set enrichment analysis web server 2016 update. Nucleic Acids Research. 2016; gkw377 .
modEnrichr
A suite of gene list enrichment analysis tools
FlyEnrichr
YeastEnrichr
WormEnrichr
FishEnrichr
Click here to raise an issue on GitHub



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21,483,193 lists analyzed
318,171 terms
156 libraries
Analyze
What's new?
Libraries
Gene search
About
Help
Upload the input list of genes
Input data
Choose an input file to upload. Either in BED format or a list of genes.
Try an example BED file.
Scegli file Nessun file selezionato
Paste a list of valid Entrez gene symbols on each row in the text-box below. Try a gene set example.
ADORA2B
AGPAT9
AIM1
AKR1C2
ALPK2
AMIGO2
ANLN
ANXA2P3
AOX1
APOBEC3B
Or
Paste the list of genes
171 gene(s) entered
Enter a brief description for the list in case you want to share it. (Optional)
Sample gene list
Give a name to your input list (optional)
Contribute
Submit
Please acknowledge Enrichr in your publications by citing the following references:
Chen EY, Tan CM, Kou Y, Duan Q, Wang Z, Meirelles GV, Clark NR, Ma'ayan A. Enrichr: interactive and collaborative HTML5 gene list enrichment analysis tool. BMC Bioinformatics. 2013;128(14).
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modEnrichr
A suite of gene list enrichment analysis tools
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YeastEnrichr
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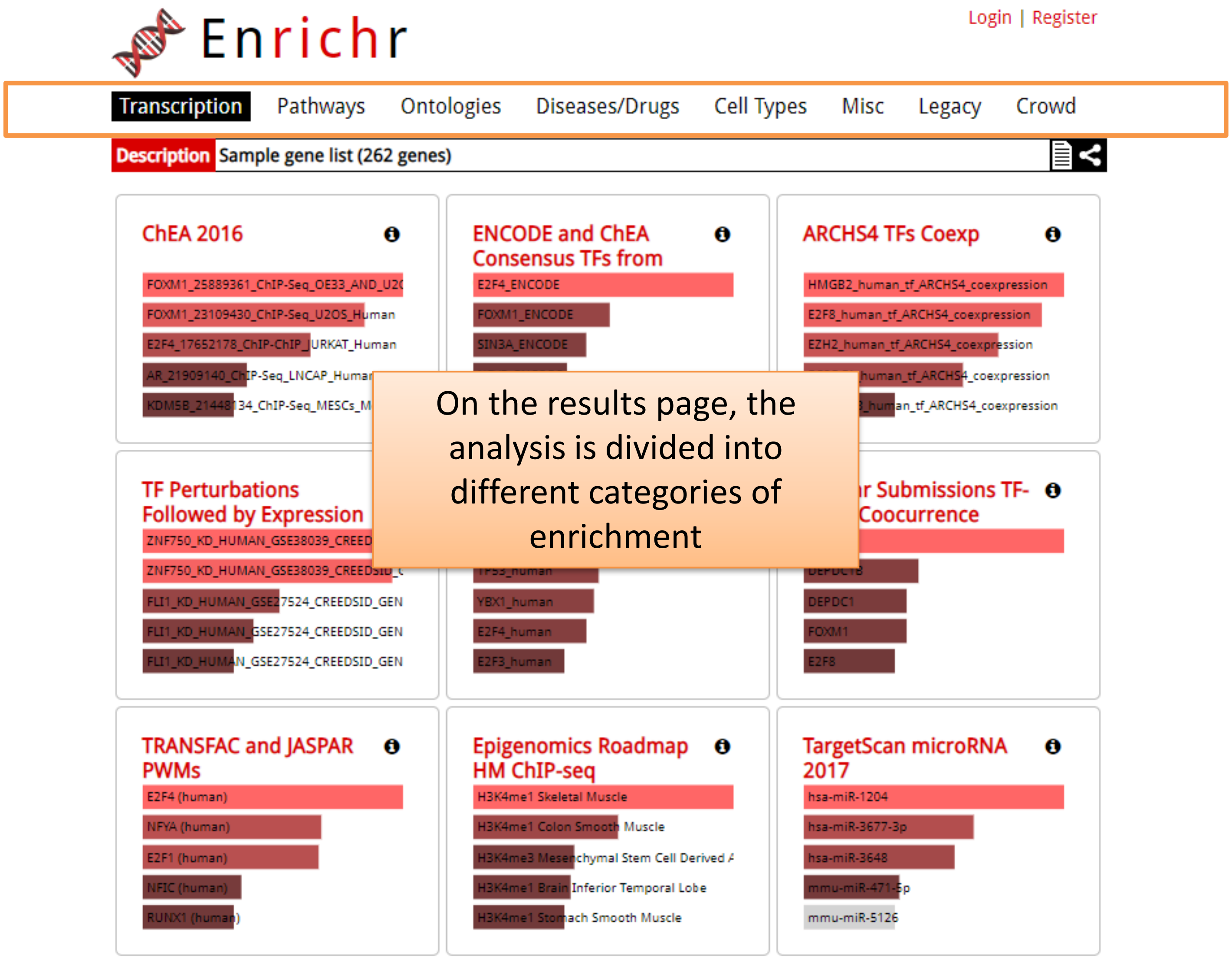
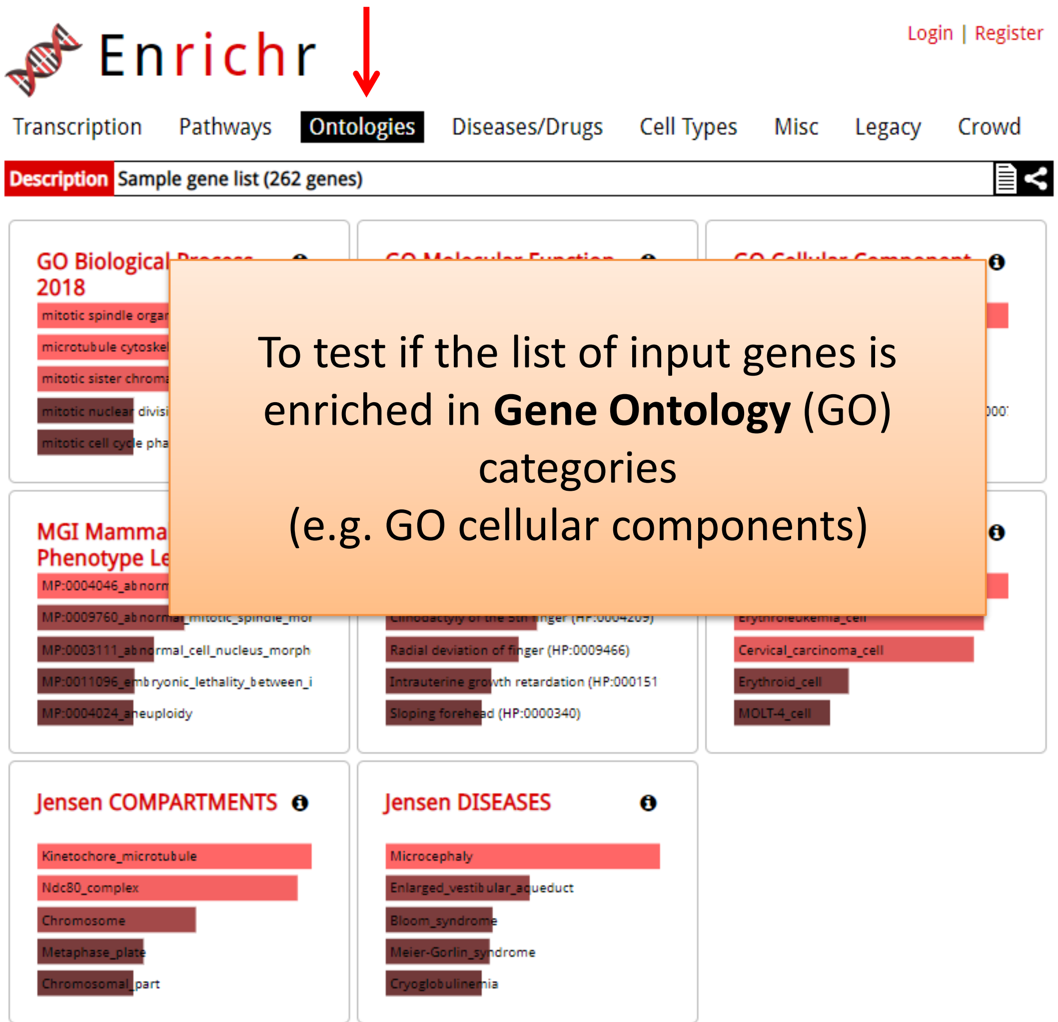
Browsing the results

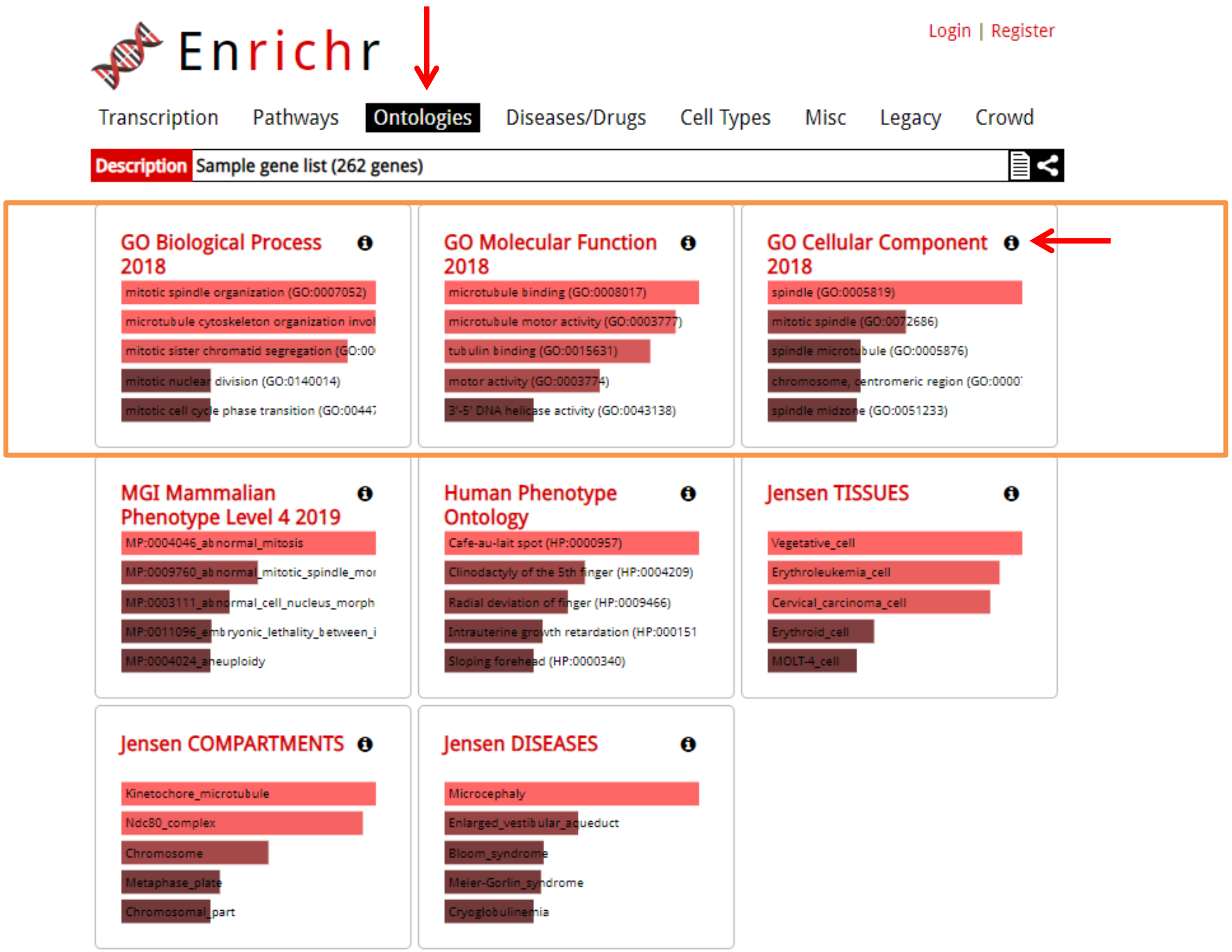


On the results page, the analysis is divided into different categories of enrichment

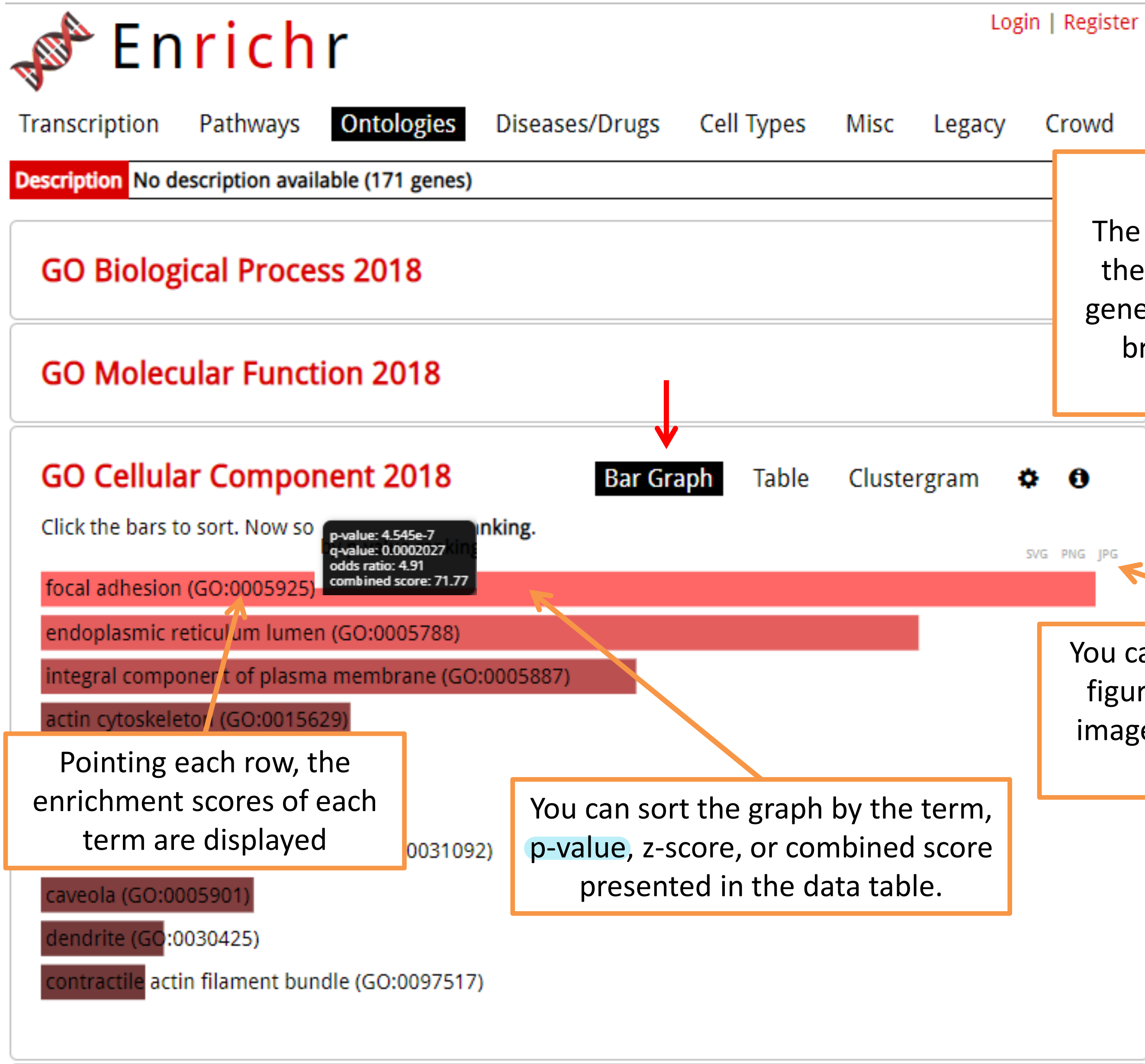
Gene Ontology (GO) results



Gene Ontology (GO)



Understanding the results



Bar graph:
The length of the bar represents the significance of that specific gene-set or term. In addition, the brighter the color, the more significant that term is.

You can export the bar graph as a figure by clicking on one of the image format buttons to the top right of the bar graph.

Pointing each row, the enrichment scores of each term are displayed

You can sort the graph by the term, p-value, z-score, or combined score presented in the data table.

Understanding the results

Enrichr

Login | Register

Transcription   Pathways   **Ontologies**   Diseases/Drugs   Cell Types   Misc   Legacy   Crowd

Description No description available (171 genes)

GO Biological Process 2018

GO Molecular Function 2018

GO Cellular Component 2018

Bar Graph   **Table**   Clustergram

Hover each row to see the overlapping genes.

10 entries per page

Enrichment scores

Enriched terms

Index	Name	P-value	Adjusted p-value	Odds Ratio	Combined score
1	focal adhesion (GO:0005925)	4.545e-7	0.0002027	4.91	71.77
2	platelet alpha granule membrane (GO:0031092)	0.009085	0.5788	13.76	64.69
3	endoplasmic reticulum lumen (GO:0005788)	0.000003915	0.0008730	5.18	64.48
4	FACIT collagen trimer (GO:0005593)	0.05022	1.000	19.49	58.31
5	protein kinase complex (GO:1902911)	0.05022	1.000	19.49	58.31
6	multivesicular body membrane (GO:0032585)	0.05834	1.000	16.71	47.48
7	microtubule minus-end (GO:0036449)	0.06640	1.000	14.62	39.65
8	myosin filament (GO:0032982)	0.08230	1.000	11.70	29.21
9	caveola (GO:0005901)	0.01282	0.7147	6.16	26.82
10	cell cortex region (GO:0099738)	0.09015	1.000	10.63	25.58

Showing 1 to 10 of 114 entries | Export entries to table   Previous   Next

Terms marked with an * have an overlap of less than 5

- **p-value (p):** computed by using the Fisher's exact test (hypergeometric test)
- **adjusted p-value:** computed by using the Benjamini-Hochberg method (FDR) for correction for multiple hypotheses testing.
- **Odds ratio (z)**: zscore computed with respect to precomputed distribution for many random input gene lists for each term in the gene set library.
- **Combined score:**

$$c = |\ln(p) * z|$$

$$\frac{k/n}{x/N}$$

Understanding the results

Enrichr

Login | Register

Transcription | Pathways | **Ontologies** | Diseases/Drugs | Cell Types | Misc | Legacy | Crowd

Description No description available (171 genes)

GO Biological Process 2018

GO Molecular Function 2018

GO Cellular Component 2018

Bar Graph | **Table** | Clustergram

Hover each row to see the overlapping genes.

10 entries per page

Search:

SVIL, MME, ITGA3, CAV1, FHL2, PLAUR, RRAS2, NEXN, CNN2, PROCR, DAB2, MYH9, FLNB, ITGA5, EPHA2

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3	endoplasmic reticulum lumen (GO:0005788)	0.000003915	0.0008730	5.18	64.48
4	FA[illegible]	[illegible]	1.000	19.49	58.31
5	pro[illegible]	[illegible]	1.000	19.49	58.31
6	mu[illegible] (GO[illegible])	[illegible]	1.000	16.71	47.48
7	microtubule minus-end (GO:0036449)	0.06640	1.000	14.62	39.65
8	myosin filament (GO:0032982)	0.08230	1.000	11.70	29.21
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Showing 1 to 10 of 114 entries | Export entries to table

Previous Next

Terms marked with an * have an overlap of less than 5

Pointing each row, the input genes related to that term are displayed

The list of input genes will be statistically enriched in terms with p-value < 0.05

Understanding the results

you can filter the results by searching for a specific term

GO Cellular Component 2018

Bar Graph | **Table** | Clustergram

Hover each row to see the overlapping genes.

10 entries per page

Search:

Index	Name	P-value	Adjusted p-value	Odds Ratio	Combined score
1	focal adhesion (GO:0005925)	4.545e-7	0.0002027	4.91	71.77
2	platelet alpha granule membrane (GO:0031092)	0.009085	0.5788	[illegible]	[illegible]
3	endoplasmic reticulum lumen (GO:0005788)	0.000003915	0.0008730	[illegible]	[illegible]
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you can sort the table by the term, p-value, z-score, or combined score

Showing 1 to 10 of 114 entries | Export entries to table

Terms marked with an * have an overlap of less than 5

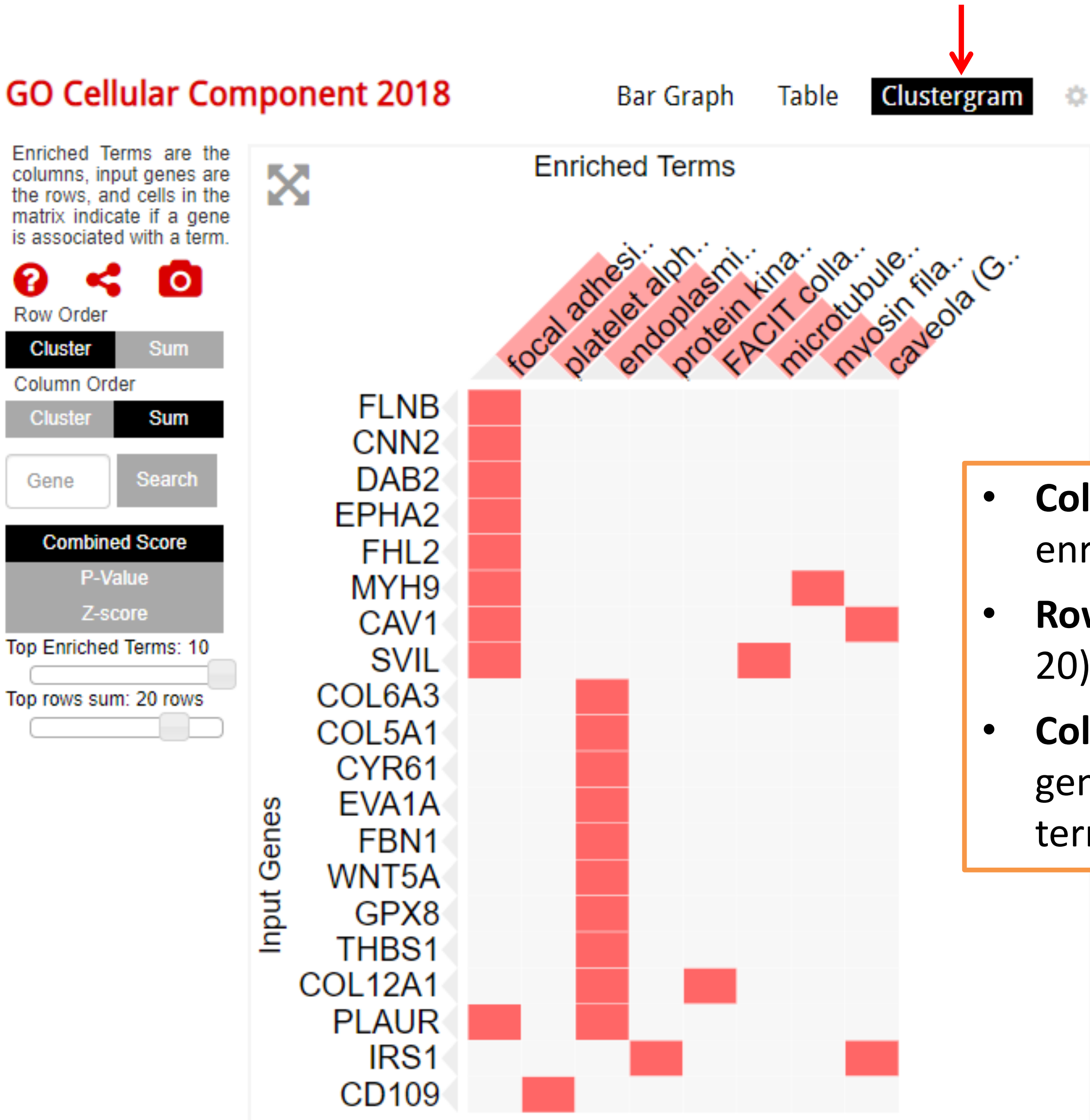
…ous Next

You can also get the table information in tab-delimited format by clicking on the "Export entries to table" button.

Exporting TABLE

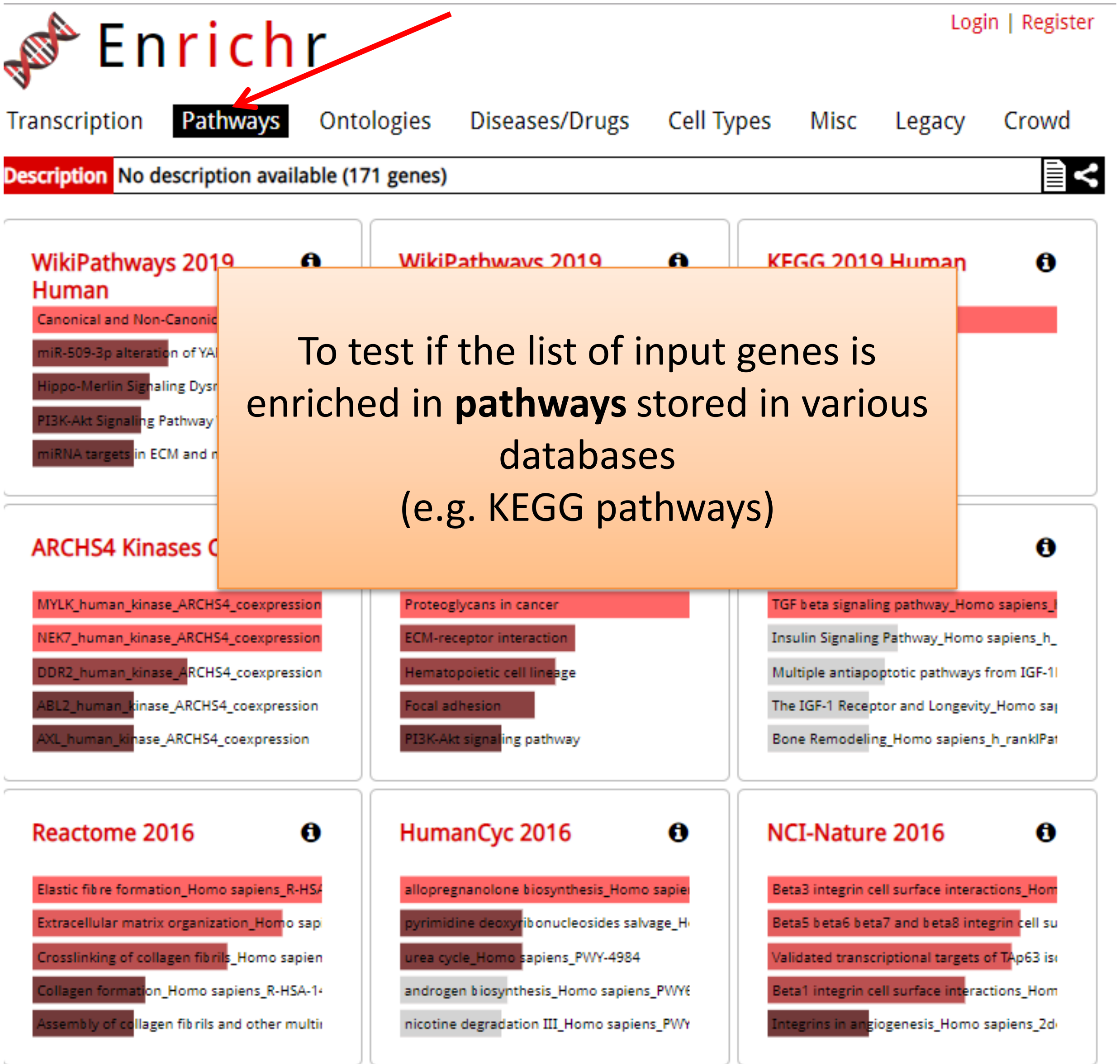
Functional category — **pvalue** — **Adjusted pvalue** — **Genes of your input list annotated in that category**

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Term	Overlap	P-value	Adjusted P-value	Old P-valu	Old Adju	Odds Ratio	Combined S	Genes					
2	focal adhesion (GO:0005925)	15/357	4.54E-07	2.03E-04	0	0	4.9142464	71.7684844	SVIL;MME;ITGA3;CAV1;FHL2;PLAUR;RRAS2;NEXN;CNN2;PROCR;DAB2;MYH9;FL					
3	endoplasmic reticulum lumen (GO:0005788)	12/271	3.91E-06	8.73E-04	0	0	5.178999158	64.4824702	EVA1A;COL5A1;STC2;COL12A1;WNT5A;GPX8;PLAUR;COL6A3;PRSS23;THBS1;CYR					
4	integral component of plasma membrane (GO:0005887)	27/1464	1.22E-04	0.018124865	0	0	2.157031924	19.4395539	PTGFR;TENM3;IRS1;SGMS2;GPR176;SLC16A7;STXBP5;CD55;MICA;IL13RA1;MME;					
5	actin cytoskeleton (GO:0015629)	8/295	0.003951	0.440583917	0	0	3.171771236	17.5515637	ANLN;SVIL;SMTN;MLPH;CDC42EP3;PAWR;MYH9;FLNB					
6	cytoskeleton (GO:0005856)	11/521	0.0053	0.472745721	0	0	2.469385235	12.9397729	CNN2;ANLN;SVIL;SMTN;MVP;PLK1;CDC42EP3;MYH9;FLNB;HMMR;SKA1					
7	platelet alpha granule (GO:0031091)	apr-91	0.007746	0.575752103	0	0	5.141057773	24.9888186	CD109;PHACTR2;VEGFC;THBS1					
8	platelet alpha granule membrane (GO:0031092)	feb-17	0.009085	0.578823918	0	0	13.75988992	64.6875234	CD109;PHACTR2					
9	caveola (GO:0005901)	mar-57	0.01282	0.714725707	0	0	6.155740228	26.8189218	IRS1;CAV1;MYOF					
10	dendrite (GO:0030425)	5/216	0.038295	1	0	0	2.707385748	8.83264913	MLPH;MME;BDNF;PLK2;RAB27A					
11	contractile actin filament bundle (GO:0097517)	feb-41	0.047948	1	0	0	5.705320211	17.3307355	CNN2;MYH9					
12	stress fiber (GO:0001725)	feb-41	0.047948	1	0	0	5.705320211	17.3307355	CNN2;MYH9					
13	FACIT collagen trimer (GO:0005593)	01-giu	0.050222	1	0	0	19.49317739	58.3100639	COL12A1					



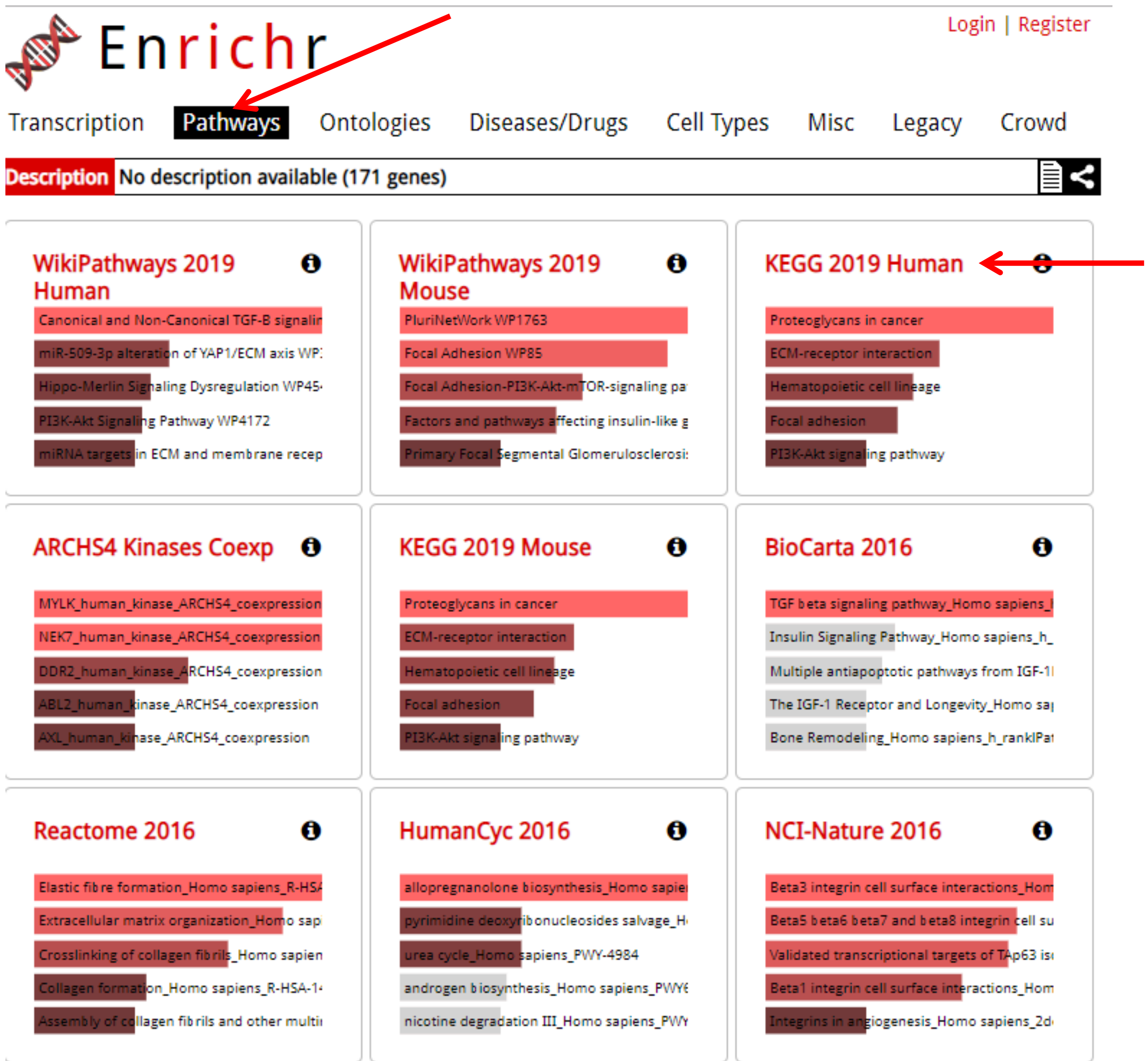
- **Columns: top-10** enriched terms
- **Rows:** input genes (first 20)
- **Colored cells:** indicate if a gene is associated with term

Pathways



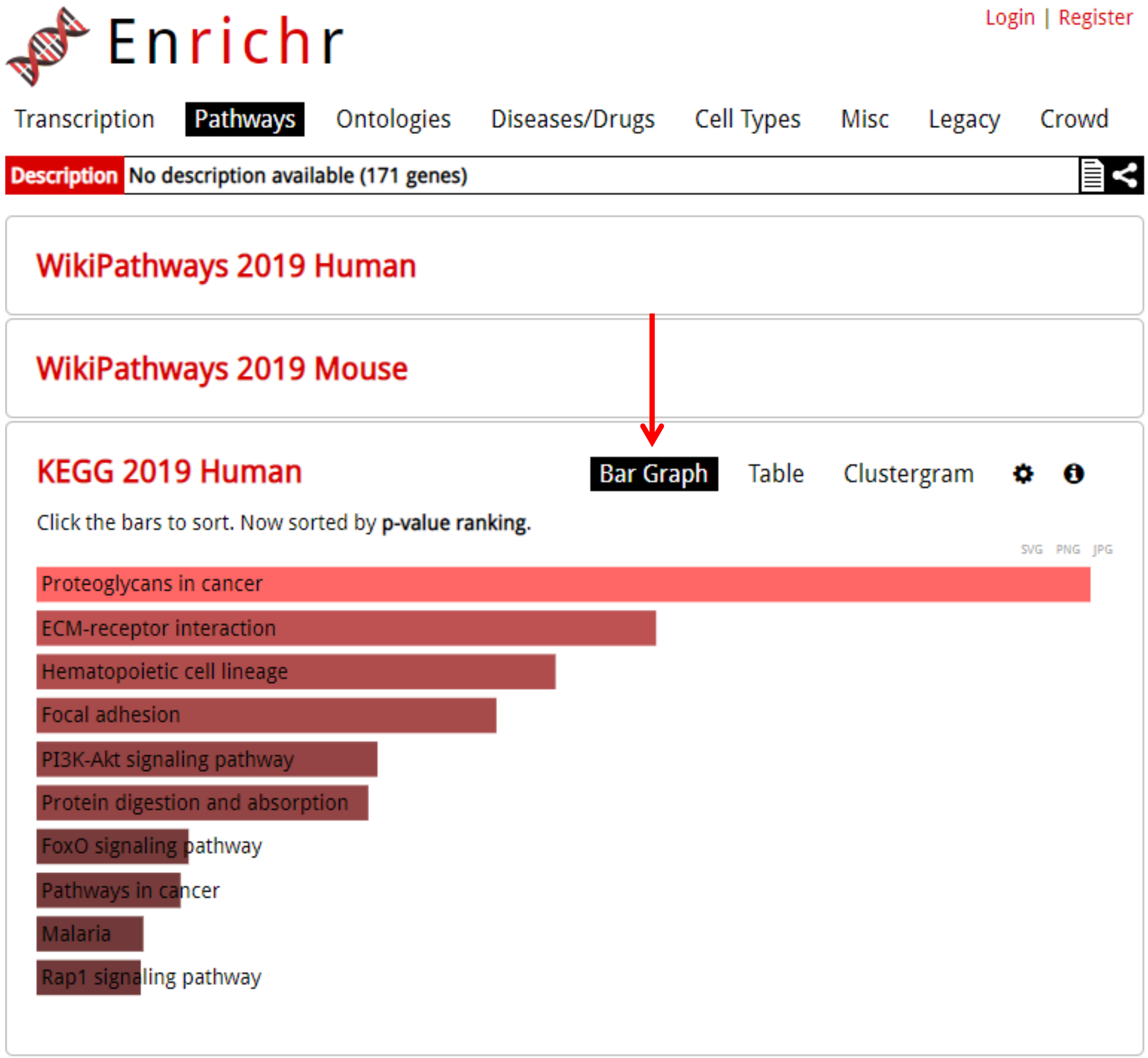
To test if the list of input genes is enriched in **pathways** stored in various databases
(e.g. KEGG pathways)

Pathways

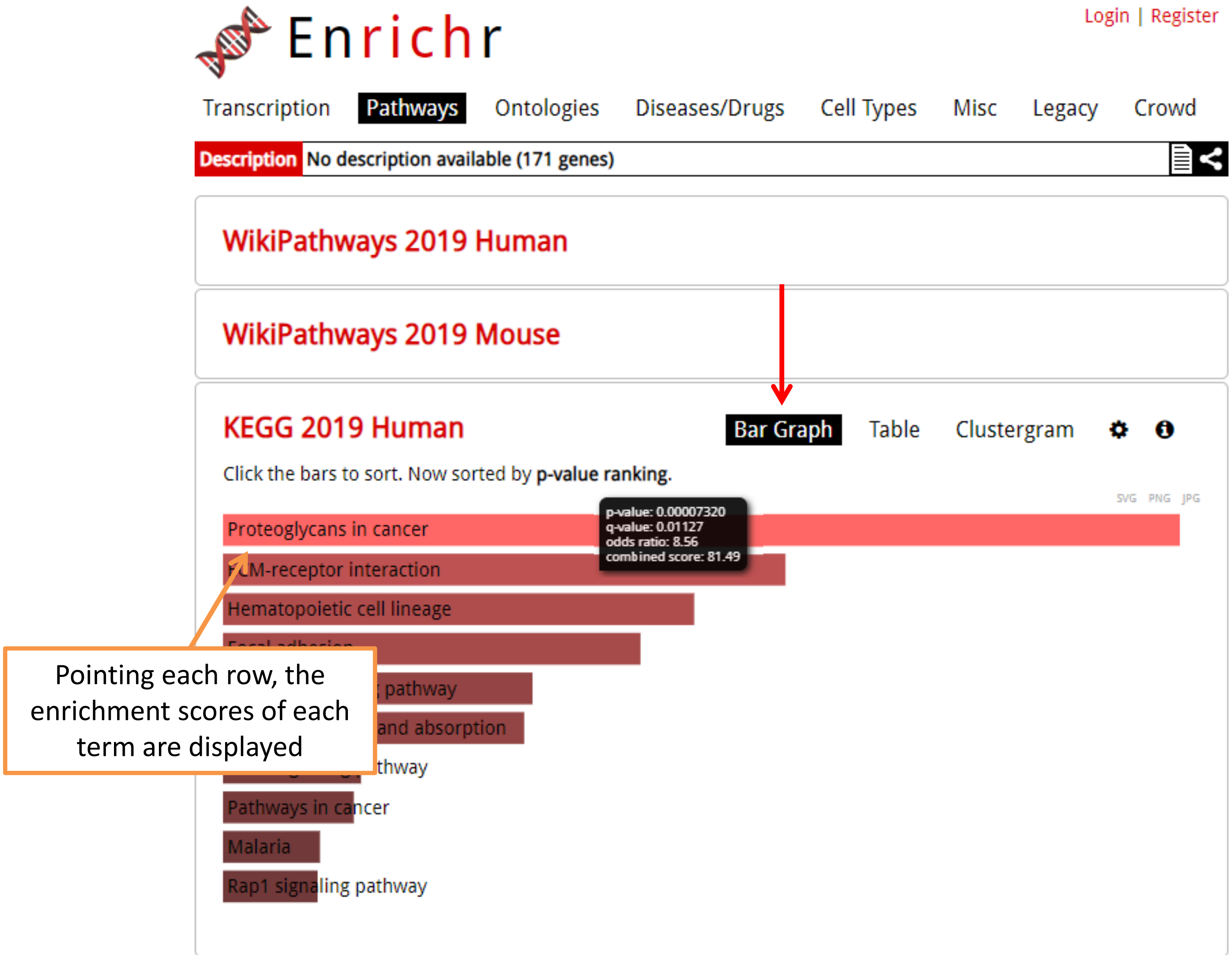


Enrichr
Login | Register
Transcription
Pathways
Ontologies
Diseases/Drugs
Cell Types
Misc
Legacy
Crowd
Description
No description available (171 genes)
WikiPathways 2019 Human
Canonical and Non-Canonical TGF-B signalir
miR-509-3p alteration of YAP1/ECM axis WP:
Hippo-Merlin Signaling Dysregulation WP45
PI3K-Akt Signaling Pathway WP4172
miRNA targets in ECM and membrane recep
WikiPathways 2019 Mouse
PluriNetWork WP1763
Focal Adhesion WP85
Focal Adhesion-PI3K-Akt-mTOR-signaling pa
Factors and pathways affecting insulin-like g
Primary Focal Segmental Glomerulosclerosi:
KEGG 2019 Human
Proteoglycans in cancer
ECM-receptor interaction
Hematopoietic cell lineage
Focal adhesion
PI3K-Akt signaling pathway
ARCHS4 Kinases Coexp
MYLK_human_kinase_ARCHS4_coexpression
NEK7_human_kinase_ARCHS4_coexpression
DDR2_human_kinase_ARCHS4_coexpression
ABL2_human_kinase_ARCHS4_coexpression
AXL_human_kinase_ARCHS4_coexpression
KEGG 2019 Mouse
Proteoglycans in cancer
ECM-receptor interaction
Hematopoietic cell lineage
Focal adhesion
PI3K-Akt signaling pathway
BioCarta 2016
TGF beta signaling pathway_Homo sapiens_l
Insulin Signaling Pathway_Homo sapiens_h_
Multiple antiapoptotic pathways from IGF-1I
The IGF-1 Receptor and Longevity_Homo saj
Bone Remodeling_Homo sapiens_h_ranklPat
Reactome 2016
Elastic fibre formation_Homo sapiens_R-HSA
Extracellular matrix organization_Homo sap
Crosslinking of collagen fibrils_Homo sapien
Collagen formation_Homo sapiens_R-HSA-1
Assembly of collagen fibrils and other multi
HumanCyc 2016
allopregnanolone biosynthesis_Homo sapie
pyrimidine deoxyribonucleosides salvage_H
urea cycle_Homo sapiens_PWY-4984
androgen biosynthesis_Homo sapiens_PWY
nicotine degradation III_Homo sapiens_PWY
NCI-Nature 2016
Beta3 integrin cell surface interactions_Hom
Beta5 beta6 beta7 and beta8 integrin cell su
Validated transcriptional targets of TAp63 is
Beta1 integrin cell surface interactions_Hom
Integrins in angiogenesis_Homo sapiens_2d

Understanding the results



Understanding the results



Understanding the results

Enrichr

Login | Register

Transcription | **Pathways** | Ontologies | Diseases/Drugs | Cell Types | Misc | Legacy | Crowd

Description No description available (171 genes)

WikiPathways 2019 Human

WikiPathways 2019 Mouse

KEGG 2019 Human

Bar Graph | **Table** | Clustergram

Hover each row to see the overlapping genes.

10 entries per page

Search:

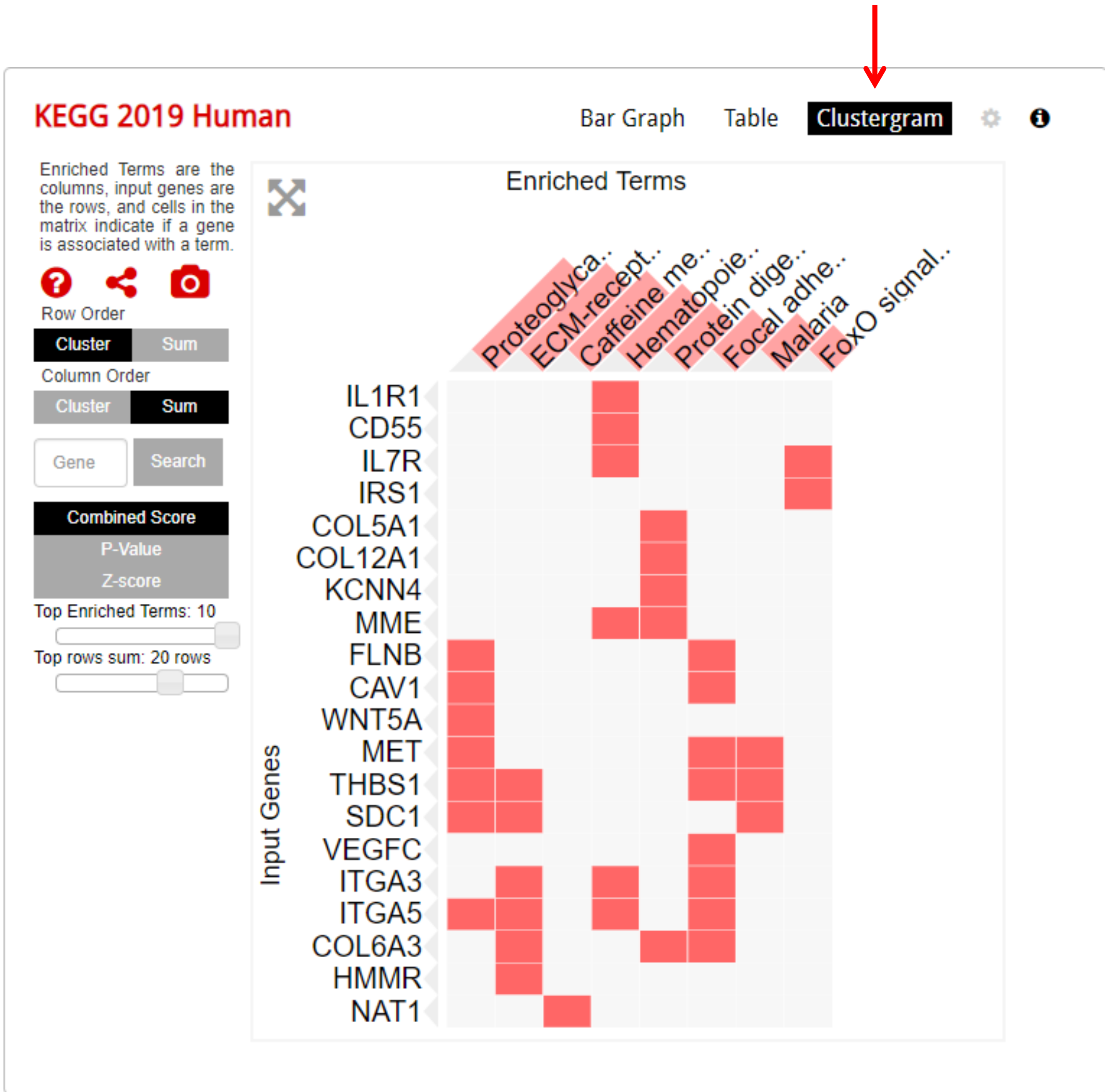
Index	Name	P-value	Adjusted p-value	Odds Ratio	Combined score
1	Proteoglycans in cancer	0.000001306	0.0004023	6.40	86.72
2	ECM-receptor interaction	0.00007320	0.01127	8.56	81.49
3	Caffeine metabolism	0.04203	0.5884	23.39	74.14
4	Hematopoietic cell lineage	0.0001859	0.01909	7.23	62.15
5	Vitamin B6 metabolism	0.05022	0.6187	19.49	58.31
6	Protein digestion and absorption	0.001054	0.05411	6.50	44.54
7	Focal adhesion	0.0003216	0.02477	4.70	37.81
8	Malaria	0.008483	0.2903	7.16	34.15
9	Ubiquinone and other terpenoid-quinone biosynthesis	0.09015	0.8167	10.63	25.58
10	FoxO signaling pathway	0.005572	0.2452	4.43	22.99

Showing 1 to 10 of 169 entries | Export entries to table

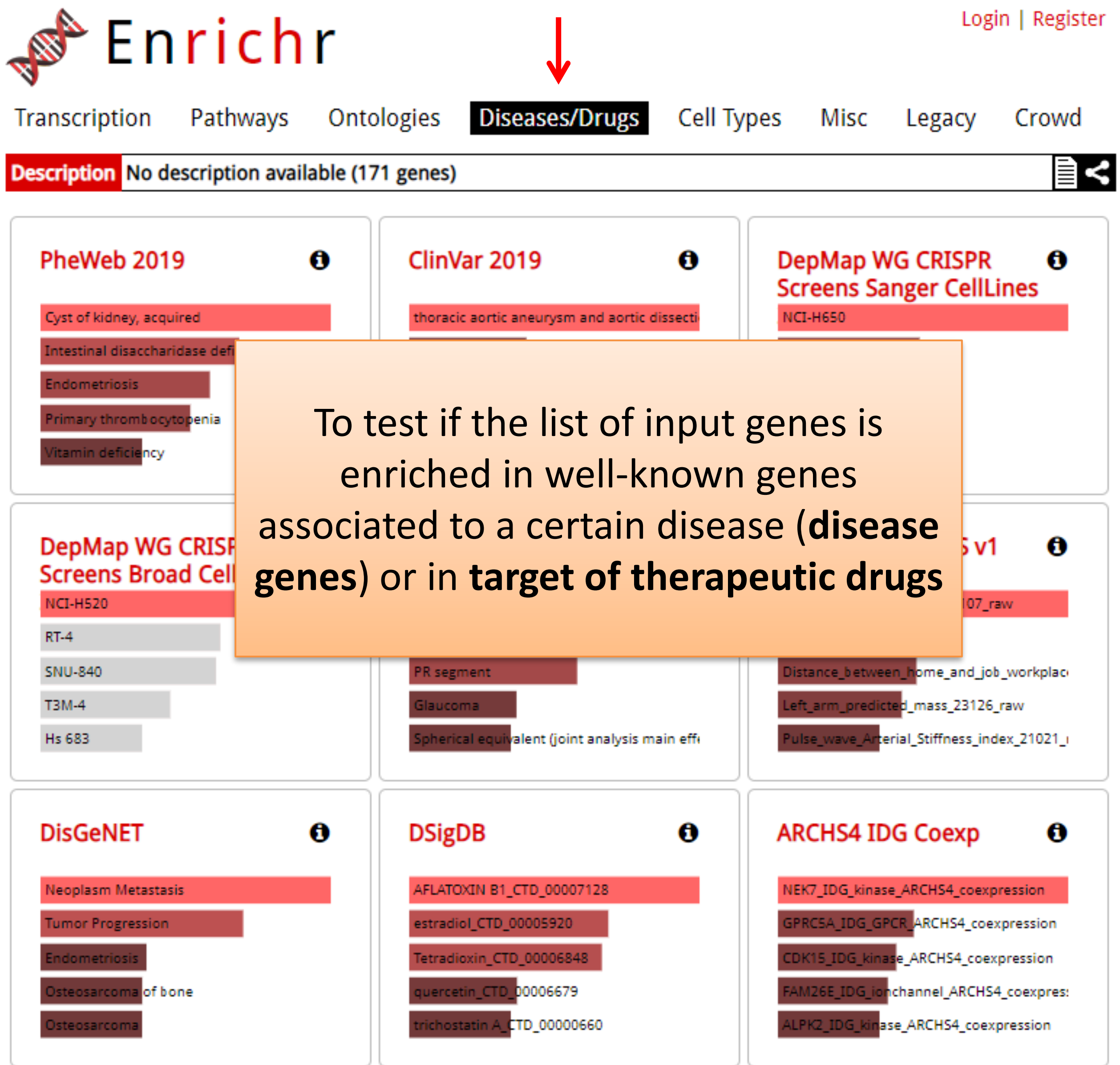
Previous Next

Terms marked with an * have an overlap of less than 5

Understanding the results

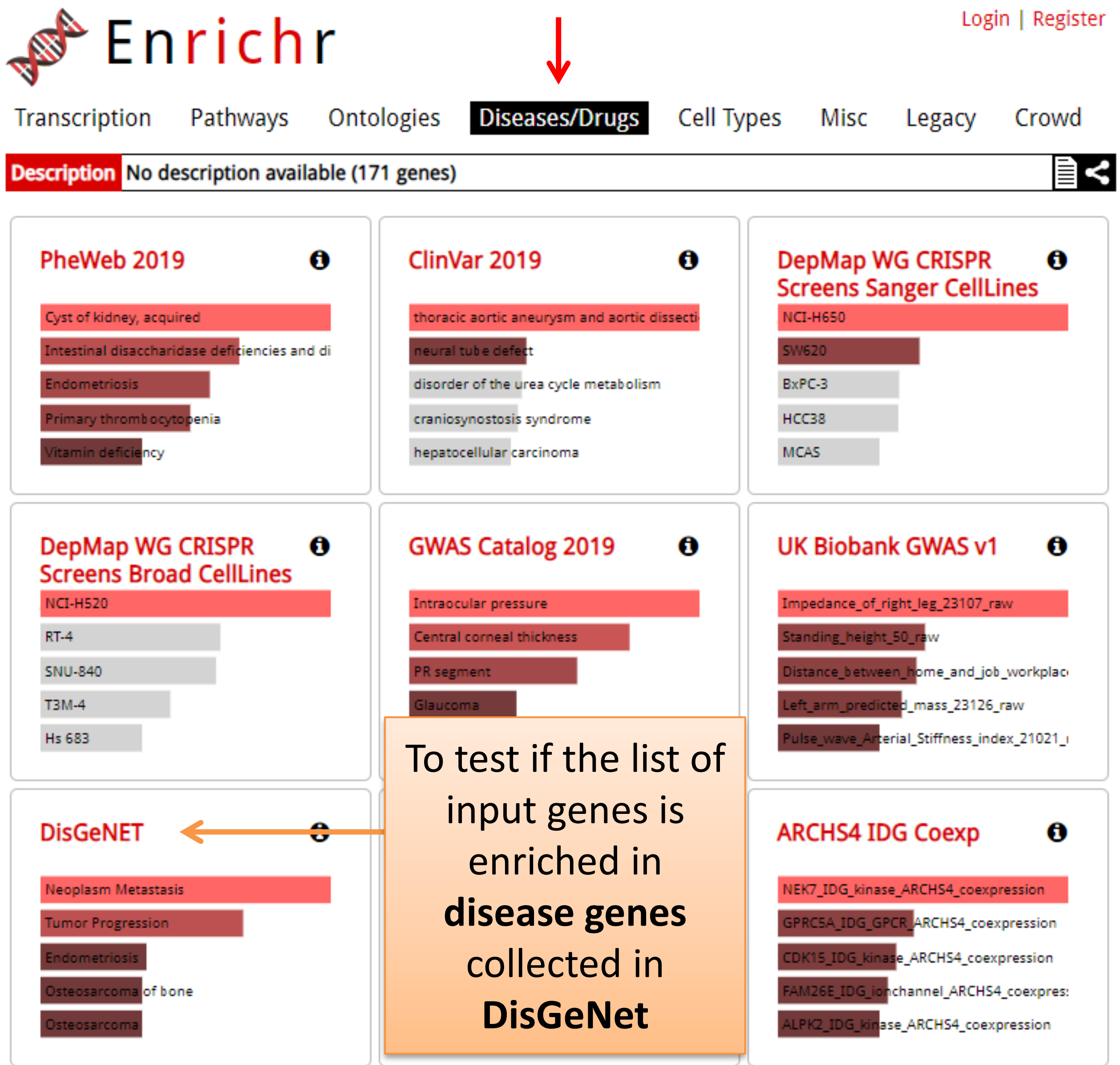


Disease/Drugs



To test if the list of input genes is enriched in well-known genes associated to a certain disease (**disease genes**) or in **target of therapeutic drugs**

Disease/Drugs



To test if the list of input genes is enriched in **disease genes** collected in **DisGeNet**

Enrichr

Login | Register

Transcription Pathways Ontologies Diseases/Drugs Cell Types Misc Legacy Crowd

Description No description available (171 genes)

PheWeb 2019

ClinVar 2019

DepMap WG CRISPR Screens Sanger CellLines 2019

DepMap WG CRISPR Screens Broad CellLines 2019

GWAS Catalog 2019

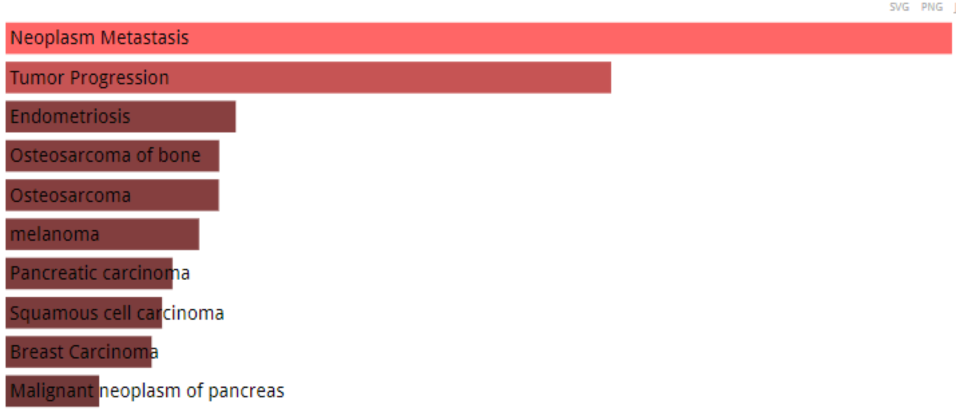
UK Biobank GWAS v1

DisGeNET

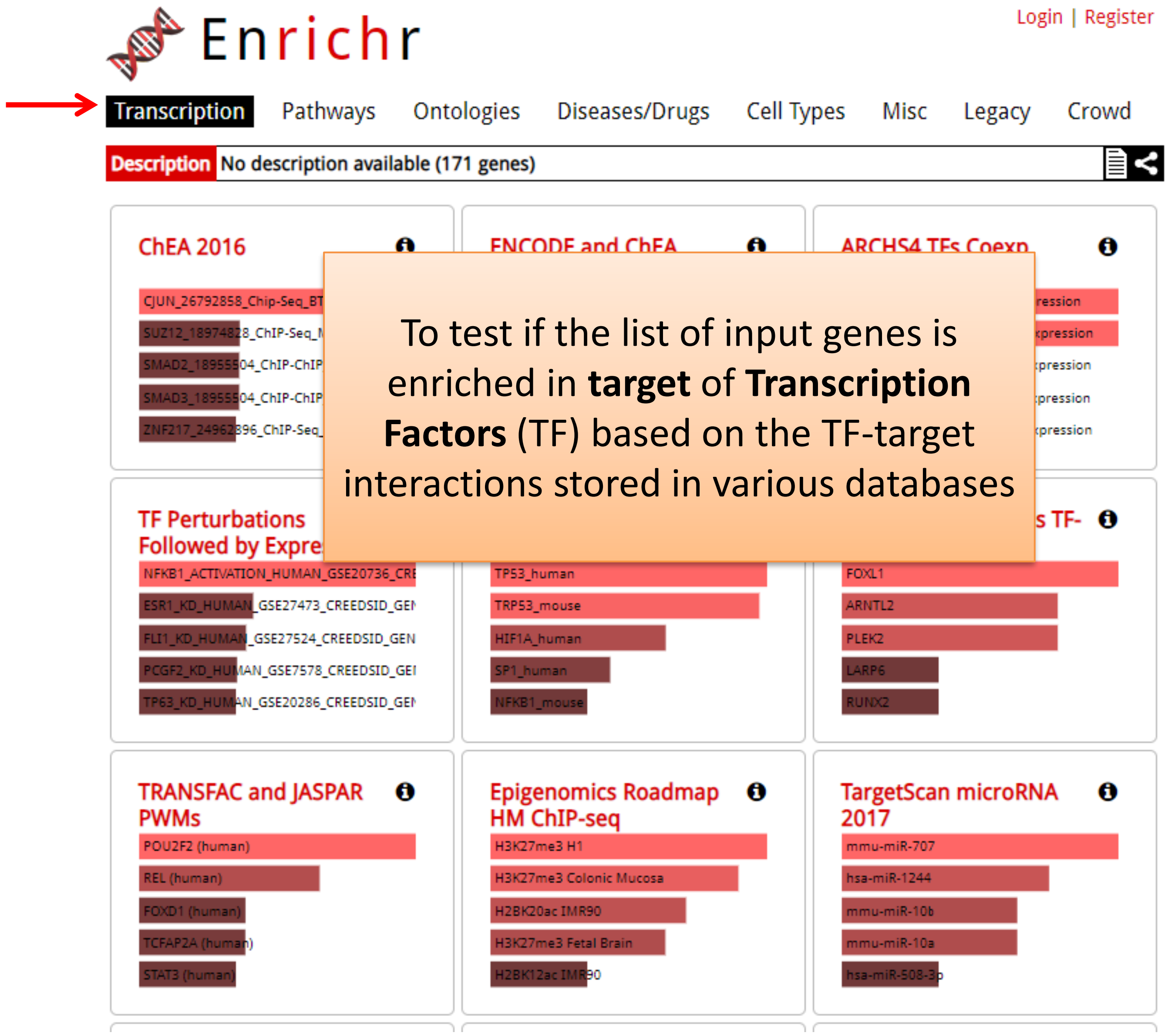
Bar Graph Table Clustergram

Click the bars to sort. Now sorted by **p-value ranking**.

SVG PNG JPG

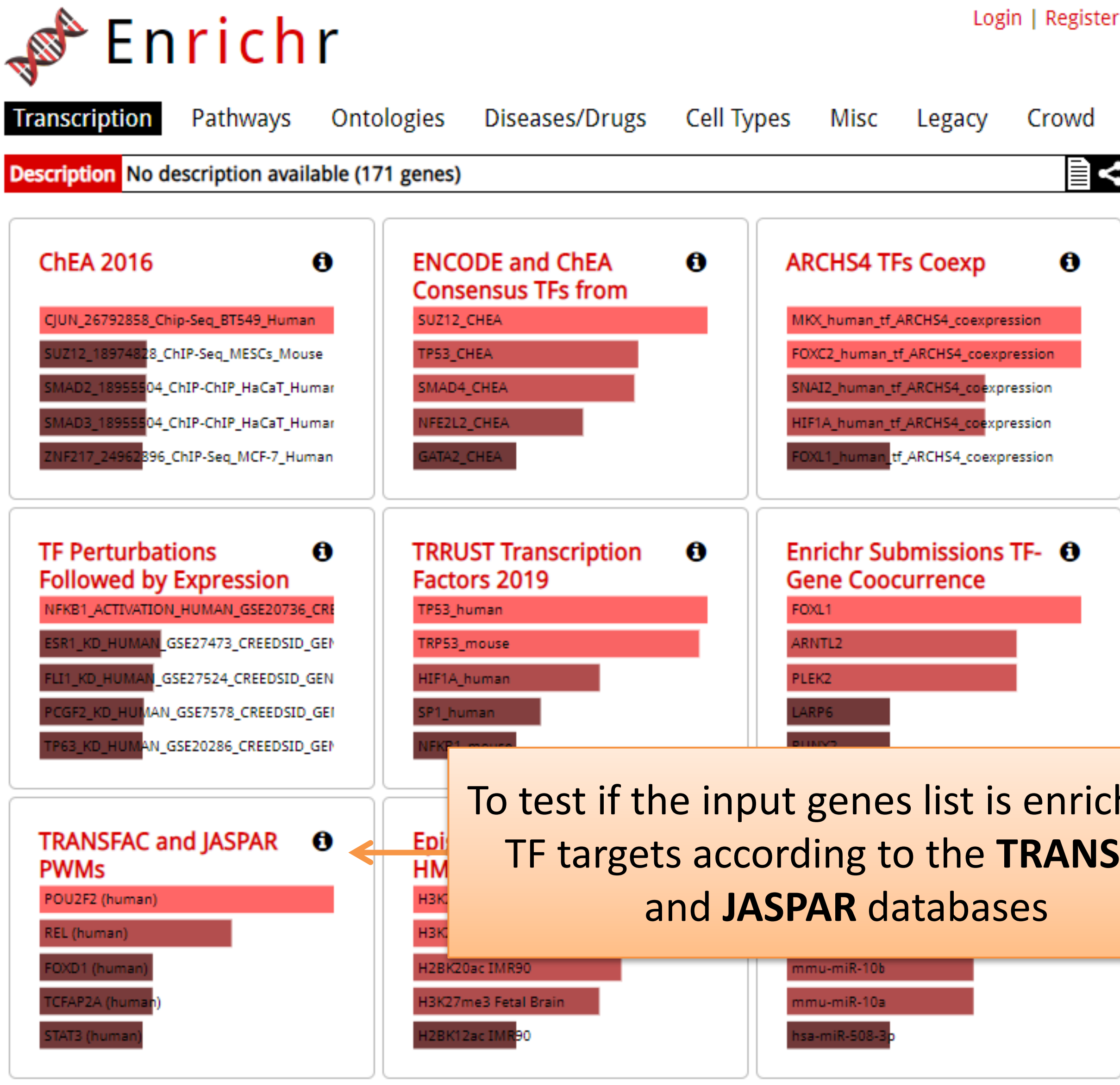


Transcription



To test if the list of input genes is enriched in **target** of **Transcription Factors** (TF) based on the TF-target interactions stored in various databases

Transcription



To test if the input genes list is enriched in TF targets according to the **TRANSFAC** and **JASPAR** databases

Transcription

TRANSFAC and JASPAR PWMs

Bar Graph **Table** Grid Network Clustergram

Hover each row to see the overlapping g

10 ▾ entries per page

TCF19, TRAM2

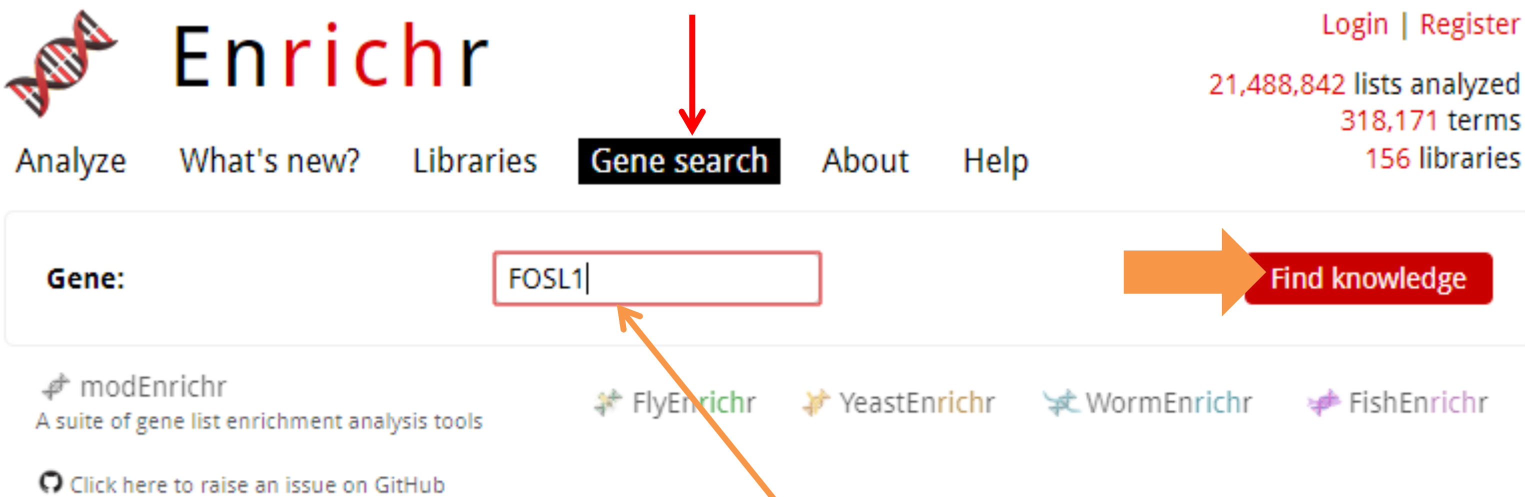
Pointing on the TF name, you can see the names of your input genes list whose promoter regions have a consensus sequence for the TF binding

Index	Name				ined score
1	MYC::MAX (human)				20.38
2	REL (human)				11.91
3	FOXD1 (human)				11.26
4	FOXM1 (human)	0.08808	1.000	4.03	9.80
5	POU2F2 (human)	0.003967	1.000	1.55	8.57
6	Klf4 (mouse)	0.05150	1.000	2.87	8.51
7	MIR210 (human)	0.1206	1.000	3.34	7.07
8	SMAD3 (mouse)	0.1263	1.000	3.25	6.72
9	FOXO3 (human)	0.1040	1.000	2.66	6.02
10	TCFAP2A (human)	0.02485	1.000	1.63	6.01

Showing 1 to 10 of 290 entries | Export entries to table

Previous Next

Gene search



You can search for a single gene and the categories in which it is involved (for which is annotated)

Enrichr

Login | Register

22,168,943 lists analyzed
319,681 terms
157 libraries

Analyze What's new? Libraries **Gene search** About Help

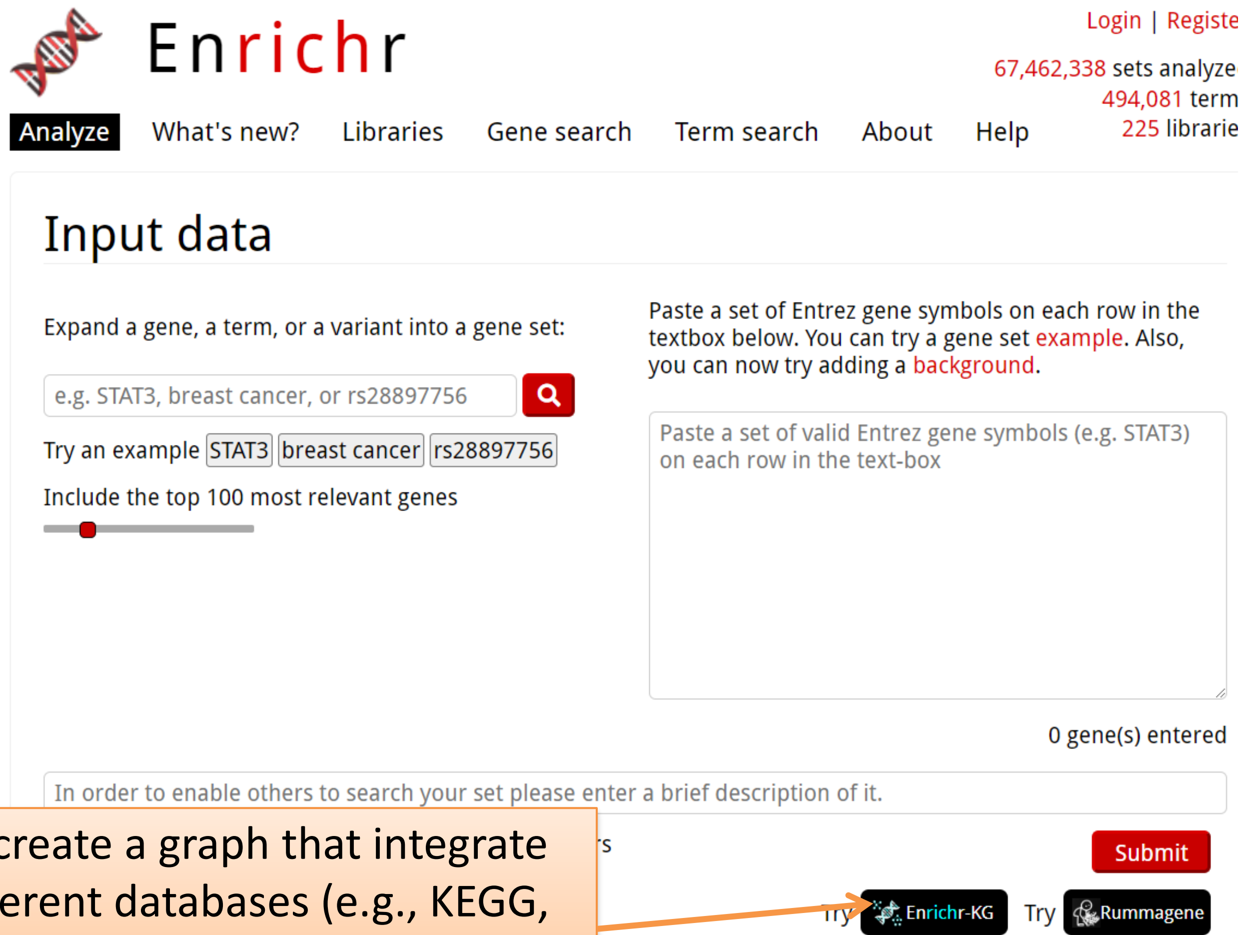
Gene: FOSL1 Find knowledge

- ⊞ **Transcription**
- ⊟ **Pathways**
  - ⊞ **BioPlanet 2019**
  - ⊞ **WikiPathways 2019 Human**
  - ⊞ **WikiPathways 2019 Mouse**
  - ⊟ **KEGG 2019 Human**
    - **FOSL1** is a member of the **Human T-cell leukemia virus 1 infection** pathway.
    - **FOSL1** is a member of the **IL-17 signaling pathway** pathway.
    - **FOSL1** is a member of the **Osteoclast differentiation** pathway.
    - **FOSL1** is a member of the **Wnt signaling pathway** pathway.
  - ⊞ **ARCHS4 Kinases Coexp**
  - ⊞ **KEGG 2019 Mouse**
  - ⊞ **BioCarta 2016**
  - ⊞ **NCI-Nature 2016**
  - ⊞ **Panther 2016**
  - ⊞ **PPI Hub Proteins**
  - ⊞ **LINCS L1000 Kinase Perturbations down**
  - ⊞ **LINCS L1000 Kinase Perturbations up**
  - ⊞ **Kinase Perturbations from GEO down**
  - ⊞ **Kinase Perturbations from GEO up**
  - ⊞ **NURSA Human Endogenous Complexome**
  - ⊞ **SubCell BarCode**
- ⊞ **Ontologies**
- ⊞ **Diseases/Drugs**
- ⊞ **Cell Types**
- ⊞ **Misc**
- ⊞ **Legacy**
- ⊞ **Crowd**

modEnrichr
A suite of gene list enrichment analysis tools

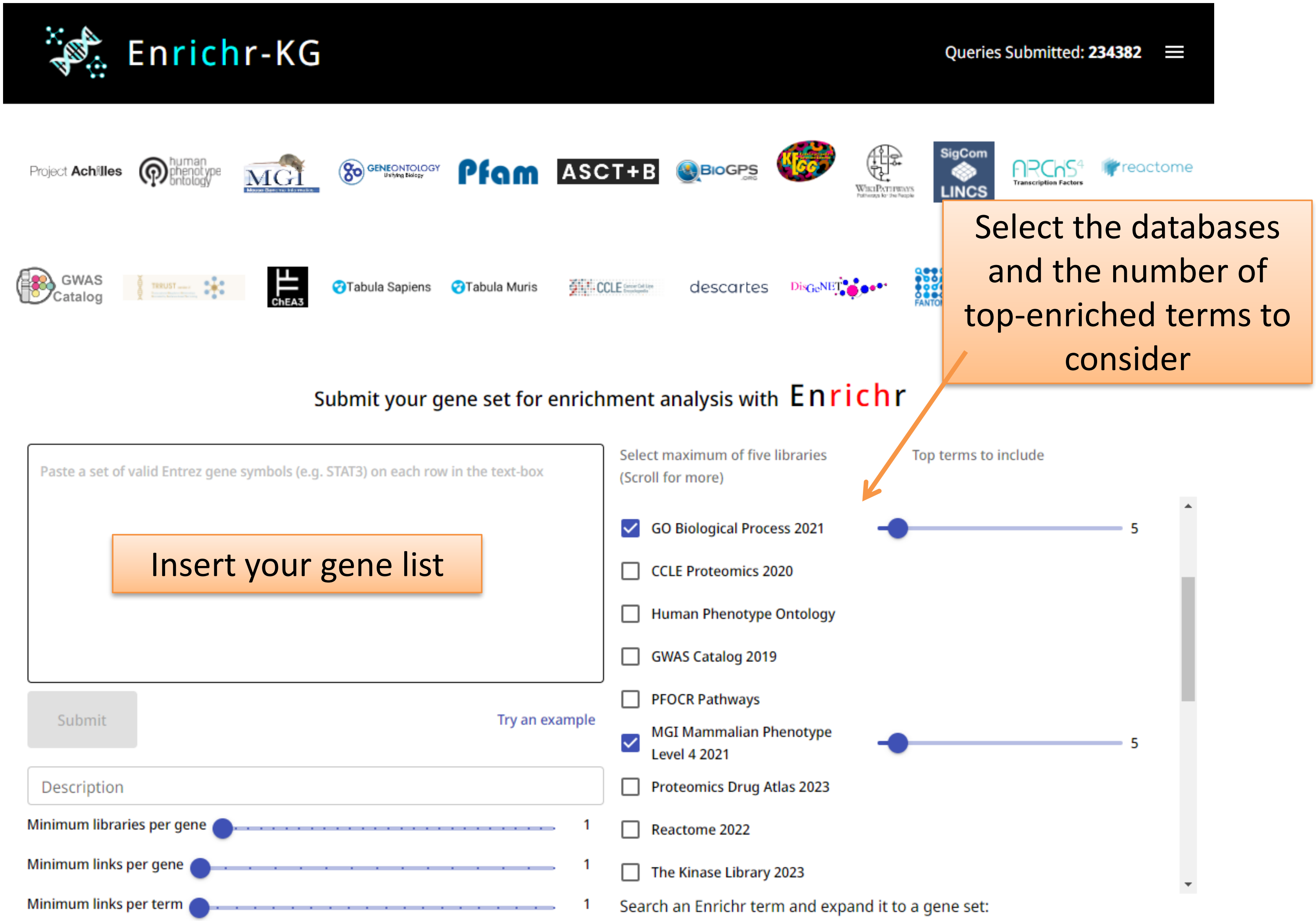
FlyEnrichr YeastEnrichr WormEnrichr FishEnrichr

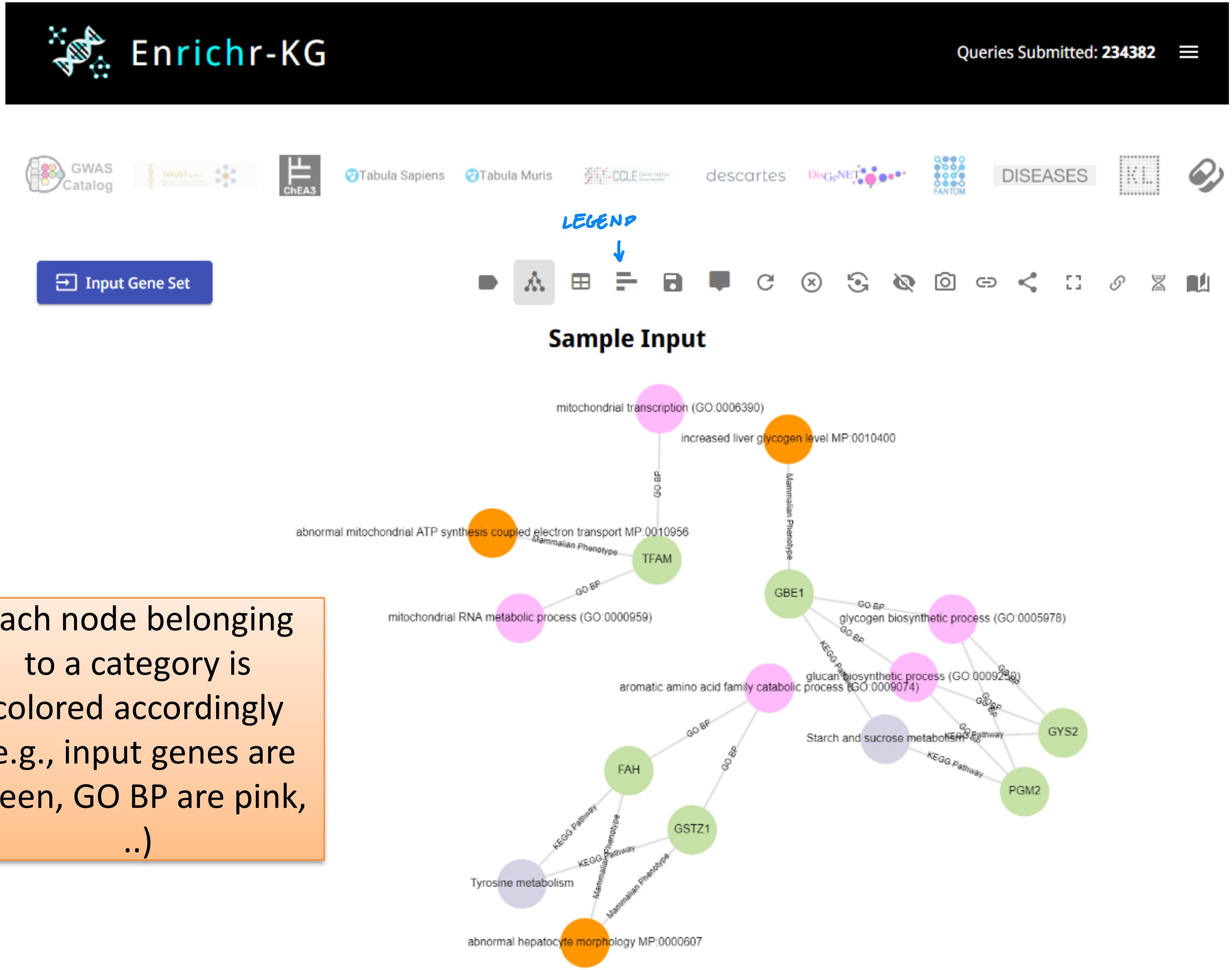
enrichR knowleged graph



To create a graph that integrate different databases (e.g., KEGG, GO) linked to the genes enriched in each one

https://maayanlab.cloud/enrichr-kg


Enrichr-KG
Queries Submitted: 234382
Project Achilles
human phenotype ontology
MGI
GENEONTOLOGY
Pfam
ASCT+B
BioGPS
KEGG
WikiPathways
SigCom LINCS
ARChS4 Transcription Factors
reactome
GWAS Catalog
TRRUST
ChEA3
Tabula Sapiens
Tabula Muris
CCLE
descartes
DisGeNET
FANTOM
Select the databases and the number of top-enriched terms to consider
Submit your gene set for enrichment analysis with Enrichr
Paste a set of valid Entrez gene symbols (e.g. STAT3) on each row in the text-box
Insert your gene list
Submit
Try an example
Description
Minimum libraries per gene
1
Minimum links per gene
1
Minimum links per term
1
Select maximum of five libraries
(Scroll for more)
Top terms to include
GO Biological Process 2021
5
CCLE Proteomics 2020
Human Phenotype Ontology
GWAS Catalog 2019
PFOCR Pathways
MGI Mammalian Phenotype Level 4 2021
5
Proteomics Drug Atlas 2023
Reactome 2022
The Kinase Library 2023
Search an Enrichr term and expand it to a gene set:



Each node belonging to a category is colored accordingly (e.g., input genes are green, GO BP are pink, ..)